

Beyond Pareto Fronts: A Leximax Universal-Regret Core for Robust Multicriteria Decision Support

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Abstract

Pareto optimality is a safety property, not a complete decision rule: an efficient set tells a decision maker (DM) which alternatives are not obviously bad, but offers no principled way to choose one. Practice fills this gap with *post hoc* devices—weighted sums, TOPSIS, compromise programming, knee selection, outranking thresholds—each of which reintroduces exactly the preference parameters the DM was unable to state. We propose the **Leximax Universal-Regret (LUR) core**, a Pareto-compatible decision rule that ranks feasible alternatives by the lexicographically sorted vector of *normalised regrets over a declared class of monotone probes*, and returns a single recommendation together with an auditable *regret certificate* identifying the most disappointed coalitions of criteria. We position LUR within robust multicriteria decision analysis. It recovers achievement scalarisation, the weighted sum, and the lexicographic maximin as special cases when the probe family is reduced to the corresponding single probe, and it relates to—but is operationally distinct from—stochastic multicriteria acceptability analysis (SMAA), robust ordinal regression, and preference robust optimisation. We prove existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based leximax rule under bounded perturbations, and a probe perturbation bound with explicit constants, and we give an adaptive correlation-clustering construction that keeps the probe count linear in the number of criteria. A replicated study (four front geometries, $m \in \{3, 5, 8, 10\}$, 30 replications, 480 instances) and a six-objective stochastic smart-grid dispatch case evaluate every method against six held-out preference families, both additive and non-additive. On worst-case (tail) held-out loss LUR is practically equivalent to the best robust scalarisations (negligible effect size, Cliff’s $|\delta| \leq 0.03$) and reduces loss by 11–47% relative to the distance-based and Monte-Carlo preference methods used in practice, while (i) resisting redundant criteria, (ii) minimising its certificate regret by construction, and (iii) admitting native stochastic and multi-stakeholder extensions. We are explicit about scope: LUR relocates preference modelling into a declared probe family rather than removing it, and does not dominate the strongest robust scalarisations on average-case loss. All code, data and scripts that regenerate every table and figure are released.

Keywords: Decision support; Multicriteria decision analysis; Robust decision making; Minimax regret; Achievement scalarising functions; Pareto efficiency.

1 Introduction

1.1 The Pareto front as a safety filter, not a decision object

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a logically impeccable *safety property*—it removes alternatives that are dominated and therefore indefensible. It is, however, *decisionally incomplete*. For m conflicting criteria the efficient set is typically large or infinite, and the decision maker (DM) is left with a cloud of efficient alternatives and no principled rule for selecting one.

Confronted with this gap, practice resorts to *post hoc* selection devices, each of which reintroduces the very preference information the DM was unable to supply: a weighted sum requires weights and cannot reach non-convex parts of the front (Miettinen, 1999); TOPSIS (Hwang and Yoon, 1981) requires a distance metric and weights; compromise programming (Zeleny, 1982) requires a metric exponent and weights; knee-point rules (Branke et al., 2008) are scale sensitive; outranking methods such as ELECTRE (Roy, 1991) and PROMETHEE (Brans and De Smet, 2016) require concordance/discordance thresholds; and hypervolume-based selection (Zitzler and Thiele, 1999) favours extremes and is costly. These methods are valid and widely used, but they require the analyst or DM to commit to additional modelling choices—weights, metrics, thresholds, exponents, or reference points—that may be difficult to justify when preference information is intentionally incomplete. The fundamental issue is that **the Pareto front is a safety filter, not a complete decision rule.**

1.2 From a front to a certificate

We argue that, for the purpose of *choosing*, the efficient set is often insufficient as a final decision object. We propose supplementing it with a *regret certificate*: a vector of normalised disappointments over the rational questions the DM might reasonably ask about the criteria, sorted from worst to best. Rather than asking “which alternatives are efficient?” we ask

which alternative is least fragile across all declared rational interpretations of the criteria?

This turns the output of the analysis from a *set* into a *certified recommendation*: a single alternative (or small indifference class) accompanied by an explanation of which questions are most disappointed and by how much.

It is crucial to note what this approach does and does not do. LUR is not parameter-free. It replaces a single preference vector by an explicit family of admissible probes, and it requires normalisation bounds and tolerance settings for numerical implementation. Its contribution is that these choices are declared, auditable, and separated from the final recommendation, not that they disappear.

1.3 The Leximax Universal-Regret core

We formalise this as the **Leximax Universal-Regret (LUR) core**. Given a feasible set and a *declared* class $\mathcal{Q}$ of admissible monotone probes (scalar “rational questions” over the normalised

criteria), LUR ranks alternatives by the lexicographically sorted vector of their normalised regrets across $\mathcal{Q}$, and returns the lexicographic minimiser together with its certificate. The construction rests on three principles: (i) *declared admissibility*— $\mathcal{Q}$ is part of the problem statement, not hidden inside a scalarisation; (ii) *Pareto compatibility*—under monotone probes a dominated alternative is never preferred to its dominator; and (iii) *direct computability*—the recommendation is obtained by a sequence of minimax-regret problems, without enumerating the efficient set.

1.4 Positioning and contribution

LUR belongs to *robust* multicriteria decision analysis. Its closest relatives are stochastic multicriteria acceptability analysis (SMAA) (Lahdelma et al., 1998; Tervonen and Figueira, 2007), which integrates acceptability over a preference distribution; robust ordinal regression (Greco et al., 2008; Figueira et al., 2009), which reasons over the set of value functions compatible with elicited holistic judgements; and preference robust optimisation (Armbruster and Delage, 2015; Hu and Mehrotra, 2015), which optimises against an ambiguity set of utility functions. LUR differs from all three in object and output: it computes the *lexicographic minimax of normalised regret* over an explicit, monotone probe family and returns an interpretable certificate; it requires no preference distribution, no elicited judgements, and no single scalarisation. We show that achievement scalarisation (Wierzbicki, 1980), the weighted sum, and the lexicographic maximin (Ogryczak, 1997) are recovered as special cases of the probe family, so LUR is a *unifying* as well as a new rule.

Our contributions are:

- **Core theoretical contribution:** We define the LUR order and its auditable certificate, proving existence, completeness, Pareto compatibility *under an explicit separation condition*, dominated-solution exclusion, and stability of the tolerance-based rule under bounded perturbations (§3, Appendix A).
- **Computational contribution:** We present an exact candidate-set algorithm based on sequential lexicographic sorts, and outline a conditional formulation for structured continuous cases (§4).
- **Probe construction:** We propose an adaptive, redundancy-aware correlation-clustering probe family with empirical redundancy tests and a perturbation bound (§5).
- **Empirical validation:** We validate LUR on a broadened benchmark suite with held-out additive and non-additive preferences, and apply it to a stochastic smart-grid dispatch case study (§7).
- **Extensions:** We outline stochastic and multi-stakeholder variants as initial extensions demonstrating the framework’s flexibility (§6).

The remainder of the paper reviews related work (§2), develops the core and its guarantees (§3), the direct formulation (§4), the adaptive probes (§5) and the extensions (§6), reports the computational study (§7), discusses limitations (§8) and concludes (§9). All proofs are in Appendix A.

2 Background and related work

2.1 Pareto efficiency and *post hoc* selection

For a vector objective $F(x) = (f_1(x), \dots, f_m(x))$ (minimisation), x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all i with at least one strict inequality; the efficient set is the non-dominated set. Because it is typically large, a single recommendation is obtained by *post hoc* selection. The weighted sum, TOPSIS (Hwang and Yoon, 1981), compromise programming (Zeleny, 1982), knee selection (Branke et al., 2008), ELECTRE (Roy, 1991), PROMETHEE (Brans and De Smet, 2016) and hypervolume selection (Zitzler and Thiele, 1999) are the standard tools; each requires weights, thresholds, a metric, or an already generated front. LUR removes these by folding the selection criterion into a declared probe family and computing the recommendation directly.

2.2 Achievement scalarising functions and ordered aggregation

Achievement scalarising functions (ASF) project a reference point onto the efficient set through a parameterised metric interpolating between L_1 and L_∞ (Wierzbicki, 1980; Miettinen and Mäkelä, 2002). Ordered weighted averaging (Yager, 1988) aggregates sorted criteria, and Ogryczak (1997); Ogryczak and Śliwiński (2006) connect ordered aggregation to lexicographic maximin. LUR *recovers* these as special cases when the probe family is reduced to a single probe: the grand-maximum probe alone gives the (reference-at-ideal) Chebyshev/ASF rule, the grand-mean probe alone the weighted sum, and the singleton family the lexicographic maximin of normalised regret. We do not claim LUR subsumes these methods with its full adaptive family—there it is a *different* aggregation paradigm. Its distinguishing features are that it sorts *normalised regrets* (dimensionless disappointments relative to each probe’s own attainable best and worst) over a *declared family of probes* and breaks ties through the whole sorted vector, and that it emits a certificate.

2.3 Robust MCDA: SMAA, robust ordinal regression, preference robust optimisation

The work closest to LUR lies in *robust* multicriteria decision analysis. **SMAA** (Lahdelma et al., 1998; Tervonen and Figueira, 2007) computes, by Monte-Carlo integration over a distribution of admissible weights (and possibly criterion values), acceptability indices describing how strongly each alternative is supported. **Robust ordinal regression** (Greco et al., 2008; Figueira et al., 2009) reasons over the entire set of additive value functions compatible with a DM’s holistic pairwise judgements, producing necessary and possible preference relations. **Preference robust optimisation** (Armbruster and Delage, 2015; Hu and Mehrotra, 2015) optimises the worst-case expected utility over an ambiguity set of utility functions. LUR shares the robust philosophy—perform well across a whole class of admissible preferences rather than commit to one—but differs in three ways. First, its admissible class is an explicit family of *monotone probes*, not a weight distribution or an elicited polyhedron. Second, its criterion is the *lexicographic minimax of normalised regret*, not an expected acceptability, a necessary/possible relation, or a single

worst-case utility. Third, its output is an interpretable *certificate* naming the binding coalitions. We note that SMAA is primarily an *exploratory* framework that informs the DM through acceptability indices rather than emitting a single choice; to obtain a comparable single-choice rule we use the standard SMAA-2 device of recommending the alternative with the highest first-rank acceptability index (ties broken by expected value), and we treat the comparison as one of decision philosophies as much as a benchmark. We likewise compare against a Savage minimax-regret (MMR) recommender over a sampled linear weight set (§7). More broadly, LUR is complementary to data-driven preference disaggregation and inverse MCDA (Doupoumis and Zopounidis, 2011), which infer preference parameters from holistic judgements: LUR instead dispenses with a fitted preference model and reasons over a declared class of questions, and could take an elicited probe family as input where such data exist.

2.4 Minimax regret and robust optimisation

Regret-based choice originates with Savage (1951) and Loomes and Sugden (1982); minimax regret seeks alternatives minimising the largest shortfall from the best attainable value. Robust optimisation (Ben-Tal and Nemirovski, 2002) and stochastic programming (Birge and Louveaux, 2011) address parameter uncertainty in the model rather than preference uncertainty in the DM. LUR extends minimax regret in three directions: it is *lexicographic* (after the worst regret, minimise the second worst, and so on), *probe-based* (regret is computed over rational coalitions of criteria, not raw criteria), and *direct* (computed without enumerating the efficient set).

2.5 Social choice and fairness

Aggregating criteria resembles social choice, but treating criteria as voters invites Condorcet cycles; LUR avoids this by aggregating *regrets over declared questions*, not votes over criteria. For multi-stakeholder settings we draw on the maximin principle of Rawls (1971) and the bargaining viewpoint of Nash (1950); we claim natural fairness properties (anonymity, monotonicity, worst-stakeholder protection), not a uniqueness theorem.

2.6 What is new relative to existing regret and robust-MCDA rules

While LUR shares motivation with several robust approaches, its mathematical object and output differ fundamentally:

- **Versus minimax regret:** LUR minimises a sorted vector of normalised probe regrets, not a single maximum regret over raw utilities or sampled weights.
- **Versus ASF/Chebyshev:** ASF commits to one scalarising function or one reference structure; LUR evaluates a declared family and certifies binding probes.
- **Versus SMAA:** SMAA integrates over a preference distribution and reports acceptability; LUR does not require a probability distribution and returns a leximax regret certificate.
- **Versus robust ordinal regression (ROR):** ROR reasons over compatible value functions after elicited comparisons; LUR can operate without holistic preference judgements.

Table 1: Comparison of LUR with representative selection and robust-MCDA methods.

Method	Main preference / model input	Requires candi-date/front?	Single rec.?	Pareto comp.	Explanation / certificate	Main limitation
Weighted sum	Weight vector	No	Yes	Yes ($w > 0$)	Low	Cannot recover non-supported points in non-convex fronts
SMAA	Distribution over preferences/weights	Usually	Not inherently (needs rule)	Usually	Acceptability indices	Exploratory; single-choice rule depends on post-processing
LUR (ours)	Declared monotone probe family; normalisation bounds; tolerance τ	Candidate-set exact	Yes (or small class)	Yes (if separating probes)	Full labelled regret certificate	Sensitive to probe-family and normalisation choices

- **Versus preference robust optimisation (PRO):** PRO optimises worst-case utility over an ambiguity set; LUR optimises lexicographically sorted normalised disappointments over interpretable monotone probes.
- **Versus ordered weighted aggregation (OWA):** LUR sorts regrets across questions, not criterion values themselves.

The novelty is not minimax regret alone, but lexicographic minimax over normalised disappointments induced by a declared monotone probe family, with an auditable certificate and an adaptive redundancy-aware probe construction.

Table 1 summarises where LUR sits relative to representative MCDA and robust methods.

3 The Leximax Universal-Regret core

3.1 Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \rightarrow \mathbb{R}^m$, $F(x) = (f_1(x), \dots, f_m(x))$, the vector objective.

Convention: All criteria f_i are assumed to be *minimised*. Maximisation objectives must have their sign reversed before normalisation. Consequently, lower normalised values r_i are better, and probes are non-decreasing functions of these costs.

For the finite case let $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ be a candidate set, e.g. an *a posteriori* approximation of the efficient set; for the continuous case (§4) A is not required. Criteria are normalised to $[0, 1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^*}{\max\{z_i^N - z_i^*, \varepsilon\}}, \quad z_i^* = \min_{x \in A} f_i(x), \quad z_i^N = \max_{x \in A} f_i(x), \quad (1)$$

with a small $\varepsilon > 0$ guarding against a zero range, so that $r_i \in [0, 1]$ attains both endpoints on A . If $z_i^N - z_i^*$ approaches zero, the normalisation floor ε prevents division by zero and effectively zeros out the relevance of that criterion. In practice, analysts should check if $z_i^N - z_i^* < 10^{-6}$; if so, the criterion is degenerate on the candidate set and should be excluded from probe generation to avoid numerical noise.

For the finite case all extrema ($z^\star, z^N, q^\star, q^w$ below) are taken over the candidate set A ; for the continuous case (§4) they are taken over $\mathcal{X}$. We use the worst attainable value over A in place of the true nadir over the efficient set, the two coincide when A contains the criterion-wise worst efficient points and otherwise the former is a conservative bound. The rule is invariant to positive affine rescaling of the criteria when the bounds are transformed consistently; it is not invariant to arbitrary monotone transformations (e.g. $f_i \mapsto \log f_i$), which change the distances that define regret—so such transformations are a modelling choice the analyst must make deliberately.

3.2 Admissible probes

A *probe* is a scalar function $q : [0, 1]^m \rightarrow \mathbb{R}$ encoding a rational decision question. The family $\mathcal{Q}$ is *declared*: it is part of the specification.

Definition 1 (Admissible probe). *A probe q is admissible if (i) it is non-decreasing in each r_i (monotonicity); (ii) it can be rescaled so that $q(\mathbf{0}) = 0$, $q(\mathbf{1}) = 1$ (normalisation); and (iii) it is L -Lipschitz in $\|\cdot\|_\infty$.*

Here $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. Monotonicity gives *weak* Pareto compatibility; strict compatibility needs a separation property (Definition 2). Canonical admissible probes are the *singletons* $q_i(r) = r_i$; the *grand mean* $q_\mu(r) = \frac{1}{m} \sum_i r_i$ (the utilitarian question); the *grand maximum* $q_{\max}(r) = \max_i r_i$ (the egalitarian/worst-criterion question); and, for any coalition $S \subseteq \{1, \dots, m\}$, the coalition mean and maximum. All have non-negative weights or are maxima, hence are monotone.

Definition 2 (Separating family). *A probe family $\mathcal{Q}$ separates criteria if for every i there is a probe $q \in \mathcal{Q}$ strictly increasing in r_i . Including all m singleton probes $q_i(r) = r_i$ is sufficient; LUR always does so.*

3.3 Disappointment and regret

For each probe $q \in \mathcal{Q}$ let $q^\star = \min_x q(r(x))$ and $q^w = \max_x q(r(x))$ (or a user-supplied worst-acceptable value). The *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^\star}{q^w - q^\star + \varepsilon} \in [0, 1], \quad (2)$$

with $D_q(x) = 0$ iff x is q -optimal and 1 iff x is q -worst.

We use the term *disappointment* for the normalised shortfall of an alternative from the best attainable value of a single given probe, and *regret certificate* for the vector collecting these probe-wise disappointments. This is regret-like but differs from classical pairwise Savage regret unless the probe family is interpreted specifically as the uncertainty set of possible true preference functions.

3.4 The lexicmax order

Collect $D(x) = (D_{q_1}(x), \dots, D_{q_K}(x))$, $K = |\mathcal{Q}|$, and sort it in *descending* order to obtain $D^\downarrow(x) = (D_{[1]}(x) \geq \dots \geq D_{[K]}(x))$. The **Leximax Universal-Regret order** is

$$x \preceq_{\text{LUR}} y \iff D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y), \quad (3)$$

i.e. x is preferred if its worst disappointment is smaller; on ties, its second worst; and so on. The *LUR solution* is $x^{\text{LUR}} \in \arg \min_x^{\text{lex}} D^\downarrow(x)$, and its *certificate* is the labelled, sorted vector $D^\downarrow(x^{\text{LUR}})$ together with the probes attaining the largest entries.

Because exact lexicographic comparison is discontinuous at ties, we use in practice—and in all guarantees below—a *tolerance-based* lexicmax order: fix $\tau > 0$ and compare $D^\downarrow(x)$ and $D^\downarrow(y)$ at the first coordinate where they differ by more than τ , treating coordinates within τ as equal. This is the rule implemented in the released code and the one for which stability holds (Theorem 3).

3.5 Foundational guarantees

Theorem 1 (Existence and completeness). *For any finite A and finite $\mathcal{Q}$, x^{LUR} exists and $\preceq_{\text{LUR}}$ is a complete preorder on A .*

Theorem 2 (Weak and strict Pareto compatibility). *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i . Then $D_q(x) \leq D_q(y)$ for all q (weak compatibility), so $x \preceq_{\text{LUR}} y$. If, in addition, $\mathcal{Q}$ separates criteria (Definition 2)—in particular if $\mathcal{Q}$ contains the singleton probes—then $D_q(x) < D_q(y)$ for some q and $x \prec_{\text{LUR}} y$.*

The separation hypothesis is necessary: with only the probe $q(r) = r_2$, the dominator $r(x) = (0, 1)$ and the dominated $r(y) = (0.1, 1)$ have equal disappointment, so monotonicity alone does not give a strict preference. LUR always includes the singletons, so it separates.

Corollary 1 (Dominated exclusion). *If x is Pareto-dominated by a feasible y and $\mathcal{Q}$ separates criteria, then x is not LUR-optimal.*

Theorem 3 (Stability of the tolerance-based rule). *Use the τ -tolerance lexicmax order. Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the winner x^{LUR} beats every rival y at the first coordinate $k(y)$ at which their sorted vectors differ by more than τ , with margin $D_{[k(y)]}(y) - D_{[k(y)]}(x^{\text{LUR}}) \geq \Delta$, and that all earlier coordinates agree within $\tau - 2\eta$. If $2\eta < \min\{\Delta - \tau, \tau\}$, the winner under $\hat{D}$ is unchanged.*

Theorem 3 is deliberately stated for the tolerance rule: exact lexicographic comparison is *not* perturbation-stable, because a winner that beats a rival only at a later coordinate after an exact earlier tie can be reversed by an arbitrarily small perturbation of the earlier coordinate. The tolerance τ absorbs exactly those near-ties. Proofs are in Appendix A. For continuous $\mathcal{X}$, existence requires compactness of $\mathcal{X}$ and continuity of each $q \circ r$; uniqueness is not guaranteed and a small indifference class may arise.

4 Exact candidate-set computation and conditional direct formulations

LUR is primarily a candidate-set selection rule, and that is the setting of our experiments. We first present the exact finite algorithm. We then develop, more tentatively, how the recommendation can in principle be computed *directly* on a continuous feasible set, replacing the usual two-stage pipeline by a single robust search. We present the continuous case as a formulation with assumptions whose large-scale empirical validation we leave to future work.

4.1 Exact algorithm for finite candidate sets

Given a discrete candidate set $A = \{x_1, \dots, x_N\}$: (1) form $F_{ij} = f_j(x_i)$; (2) normalise to R via (1); (3) evaluate $V_{iq} = q(r(x_i))$ for each probe; (4) set $q^\star = \min_i V_{iq}$, $q^w = \max_i V_{iq}$; (5) form disappointments (2); (6) sort each row descending; (7) return the lexicographically smallest row.

The worst-case complexity is $O(NKm + NK \log K + N \log N)$ when using a tolerance-based lexicographic comparison. Here $O(NKm)$ is the cost of probe evaluation, $O(NK \log K)$ is the cost of sorting the K -dimensional disappointment vector for each of the N candidates, and $O(N \log N)$ is the cost of the final lexicographic sort across candidates. The number of probes K is controlled by the construction of §5.

4.2 Cost of the anchors

Both the finite and continuous formulations need, for each probe, its attainable best $q^\star = \min_x q(r(x))$ and worst $q^w = \max_x q(r(x))$. On a candidate set these are $O(N)$ scans; on a continuous $\mathcal{X}$ each is a scalar global optimisation, so the family contributes $2K$ such subproblems—cheap relative to front enumeration when K is linear in m (§5) but not free, and we account for it explicitly.

4.3 Sequential minimax-regret formulation

For continuous or large discrete $\mathcal{X}$, solve a sequence of minimax problems. *Stage 1* minimises the worst regret,

$$\rho_1 = \min_{x \in \mathcal{X}} \max_{q \in \mathcal{Q}} D_q(x), \quad \mathcal{X}_1 = \{x : \max_q D_q(x) = \rho_1\}. \quad (4)$$

Stage $k > 1$ fixes the probes already saturated and minimises the next worst regret over the remaining ones, restricted to $\mathcal{X}_{k-1}$, until the optimal set is a singleton or all probes are exhausted. This realises the lexicographic order (3) by construction.

4.4 Exact sequential cumulative-sum formulation

Lexicographic minimisation of the sorted vector is exactly equivalent to sequentially minimising the cumulative sums of the p largest disappointments, $\sum_{k=1}^p D_{[k]}(x)$ for $p = 1, \dots, K$. Each of these is convex when the disappointments are convex and admits the standard ordered-value

linearisation (Ogryczak and Śliwiński, 2006). By solving K sequential problems, we obtain the exact LUR recommendation without needing to enumerate the Pareto front.

4.5 Single-stage heuristic surrogate

The stages can be collapsed with a strictly decreasing weight schedule. For a large constant $M > 1$,

$$\min_{x \in \mathcal{X}, \rho_1 \geq \dots \geq \rho_K \geq 0} \sum_{k=1}^K M^{-(k-1)} \rho_k \quad \text{s.t.} \quad D_{(k)}(x) \leq \rho_k, \quad (5)$$

where $D_{(k)}(x)$ denotes the k -th largest disappointment at x . This is an *exact* surrogate for the lexicographic order only under finite-gap assumptions. For general continuous values no finite M reproduces the exact order, so this single-stage form is primarily a heuristic approximation.

4.6 Smoothed continuous optimisation

For continuous $\mathcal{X} \subseteq \mathbb{R}^n$ the inner maximum is non-smooth; a log-sum-exp surrogate

$$\max_{q \in \mathcal{Q}} D_q(x) \approx \frac{1}{\tau} \log \left(\sum_{q \in \mathcal{Q}} \exp(\tau D_q(x)) \right) \quad (6)$$

is smooth and approaches the maximum as $\tau \rightarrow \infty$. A probe-active-set sequential convex program—linearise the active probes $\mathcal{Q}_k = \{q : D_q(x_k) \geq \rho_1 - \varepsilon\}$ at the current iterate, solve the resulting convex subproblem, update, and repeat—then converges to a local LUR solution under standard convexity assumptions, analogously to SQP.

4.7 Why this matters, and what we do not yet show

The DM specifies the probe family once; in the finite case LUR returns the least-fragile compromise by the exact procedure above, and in the continuous case the formulations here make a front-free computation *available* in principle. We are careful not to overclaim: our experiments operate on candidate sets, so the large-scale continuous advantage is argued, not yet demonstrated, and we flag its empirical validation as the main piece of future work. We quantify the probe-family savings in §5.

5 Adaptive, redundancy-aware probe generation

5.1 The redundancy problem

The full coalition family (mean and maximum over every non-trivial subset of criteria) has size exponential in m . Worse, real criteria are often *redundant*: several measured objectives track the same underlying concern and are strongly positively correlated. Averaging selection rules implicitly *over-weight* such a group—it is counted once per member—biasing the recommendation.

Consider a simple illustrative example: a 3-criteria problem where f_1 is cost, and f_2 and f_3 are two identical measures of environmental impact. An averaging rule will implicitly double-weight the environmental impact simply because it was measured twice. We need a probe family

that (i) is small, (ii) stays monotone, and (iii) is less sensitive to redundancy than averaging aggregation. LUR with adaptive probes identifies that f_2 and f_3 are identical (correlation 1.0) and clusters them, generating a reduced set of probes that does not repeatedly average the redundant pair.

5.2 Correlation clustering

We group *redundant* criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\text{Corr}(f_i, f_j) \geq \theta$. We chose single-linkage clustering intentionally because it is conservative regarding redundancy. Two criteria only need a single path of high correlations between them to be grouped together. This prevents the artificial slicing of a continuous underlying dimension into multiple independent criteria that would otherwise overwhelm the probe generation.

Redundancy is positive correlation: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria are genuine trade-offs and must remain separate—hence we threshold the signed correlation, not its absolute value. This yields clusters $C_1, \dots, C_c$.

5.3 The adaptive probe family

LUR always includes the canonical anchors—the m singletons, the grand mean, and the grand maximum—and adds one mean and one maximum probe *per cluster* of size > 1 :

$$\mathcal{Q} = \{q_i\}_{i=1}^m \cup \{q_\mu, q_{\max}\} \cup \bigcup_{|C_k|>1} \{q_{C_k, \text{mean}}, q_{C_k, \text{max}}\}. \quad (7)$$

A redundant group thus contributes *one* aggregate (cluster) question rather than a separate aggregate per member, so the family does not amplify a correlated group the way an averaging aggregate does, while the count stays $O(m + c)$ —linear, not exponential. We stress that this *reduces*, but does not eliminate, redundancy sensitivity: the singleton probes for the group’s members remain in $\mathcal{Q}$ (they are needed for separation, Theorem 2), so a redundant group can still influence lower-order lexicographic tie-breaking. The claim is therefore empirical—less redundancy bias than averaging methods on the tested benchmarks (§7)—not a duplication-invariance theorem. All probes are non-negative-weighted sums or maxima, hence monotone, so Pareto compatibility (Theorem 2) is preserved.

Theorem 4 (Non-negative-probe Pareto compatibility). *If every probe in $\mathcal{Q}$ is a non-negative-weighted sum or a maximum of the normalised criteria, then LUR is Pareto compatible.*

5.4 A perturbation bound (not a ranking guarantee)

We state the reduction effect as a regret-value perturbation lemma with explicit constants. We are deliberately modest: it bounds the change in individual disappointments, *not* the change in the LUR ranking or winner, and it depends on the normalisation floor ε , which is decision-critical.

Table 2: Probe-family size: full coalition family ($2(2^m - 1) - m$ probes) vs. adaptive correlation-clustering family (concave instances, $N = 400$).

m	full coalitions	adaptive probes	reduction
3	11	5	$2.2\times$
5	57	7	$8.1\times$
8	502	10	$50.2\times$
10	2,036	12	$169.7\times$
15	65,519	17	$3,854.1\times$

Lemma 1 (Disappointment perturbation). *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \leq \eta$ (hence also $|q^\star - q'^\star| \leq \eta$ and $|q^w - q'^w| \leq \eta$), and let the normalisation ranges satisfy $q^w - q^\star, q'^w - q'^\star \geq \beta > 0$. Then for every x*

$$|D_q(x) - D_{q'}(x)| \leq \frac{3\eta}{\beta},$$

with the floor $\beta \geq \varepsilon$ in the worst case. The bound is vacuous when β is of order η , and informative only when the probe’s attainable range is bounded away from zero—a regime the analyst can check directly.

Whether the reduced family admits such an η -close q' for every full-family probe is *not* established by correlation clustering in general; we therefore present Lemma 1 as a sensitivity statement and rely on the empirical redundancy study (§7) for evidence that clustering retains decision quality.

5.5 Empirical reduction

Table 2 reports the realised reduction on concave instances. The full family here is $\{q_{\text{mean},S}, q_{\text{max},S} : \emptyset \neq S \subseteq \{1, \dots, m\}\}$, of size $2(2^m - 1) - m$ (mean and max coincide on singletons); the adaptive family is linear in m , giving a nearly four-thousand -fold reduction by $m = 15$ with, as §7 shows, little loss of decision quality.

5.6 Greedy probe selection (optional)

When even the adaptive family is larger than needed, a greedy set-cover heuristic selects a minimal subfamily distinguishing all dominant pairs; for the max-coverage variant it attains a $(1 - 1/e)$ guarantee and for full coverage a $1 + \ln d$ guarantee (Nemhauser et al., 1978; Chvátal, 1979). We do not use it in the experiments, which already run on the compact adaptive family.

6 Extensions: stochastic and multi-stakeholder variants

6.1 Stochastic regret certificates

When criteria are noisy, model $f_i(x)$ as drawn from a joint distribution $P(x)$. Each probe value is then a random variable; for non-linear probes (e.g. q_{max}) its mean $\mu_q(x)$ is *not* a simple function of the criterion marginal means. We estimate the robust stochastic probe value by its

finite sample average over S scenarios $\omega^{(1)}, \dots, \omega^{(S)}$:

$$\hat{\mu}_q(x) = \frac{1}{S} \sum_{s=1}^S q(r(x, \omega^{(s)})). \quad (8)$$

The normalisation anchors q^*, q^w are taken on these estimated means. We define the α -confidence regret

$$D_q^\alpha(x) = \frac{\hat{\mu}_q(x) + z_\alpha \hat{\sigma}_q(x) - q^*}{q^w - q^* + \varepsilon}, \quad (9)$$

with z_α the standard-normal quantile, penalising both expected disappointment and its sample uncertainty $\hat{\sigma}_q(x)$, and apply the tolerance-based leximax order to $(D_{q_1}^\alpha, \dots, D_{q_K}^\alpha)$. We present this as a practical risk-adjusted rule; we note that sample-average probes preserve exact Pareto compatibility.

Proposition 1 (Sample-average stochastic consistency). *For any finite sample size S , if x strictly dominates y under all scenarios $\omega^{(s)}$, then $x \prec_{\text{LUR}} y$ under the sample-average probe regrets.*

§7 gives an empirical confirmation.

6.2 Multi-stakeholder (Rawlsian) exploratory variant

Let $\mathcal{U} = \{1, \dots, U\}$ index stakeholders, each with importance weights $w_u \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^u(x)$. An exploratory *Rawlsian LUR* applies the leximax order to the worst-stakeholder disappointments, $\max_u D_q^u(x)$. It maintains natural fairness properties such as anonymity and worst-stakeholder protection. A full treatment of this extension is beyond the scope of this paper, but we include it in our empirical evaluation to demonstrate the flexibility of the regret-certificate framework.

7 Computational study

7.1 Design and protocol

We evaluate selection *quality* under a held-out protocol: each method picks one alternative using only its own machinery, and the pick is scored against preference models that *no* method used during selection. To avoid biasing the evaluation toward the additive/scalarising logic of the probes, the held-out set spans **six** families—random linear $-w^\top r$, weighted Chebyshev $-\max_i w_i r_i$, augmented ASF, CES, a *non-additive* 2-additive Choquet integral with random non-negative Möbius masses on singletons and pairs, and a *satisficing* threshold utility penalising exceedances of random aspiration levels. Weights are drawn from a uniform simplex; 300 utilities are sampled per family per replication. We report two non-negative, lower-is-better losses, each normalised per utility against the best and worst feasible alternatives: the *mean held-out loss* and the *tail held-out loss*. CVaR_{10%} (the average loss over the worst 10% of utility draws) serves as our robust tail measure. We intentionally report both mean loss and tail loss, emphasizing that robust methods will typically underperform on average in exchange for bounding the worst-case disappointment.

Table 3: Held-out loss over four geometries $\times m \in \{3, 5, 8, 10\}$, 30 replications (480 instances; lower is better). Tail loss is the worst-case (CVaR_{10%}) metric. Best in each column in **bold**; on tail loss LUR is within a negligible effect size ($|\delta| \leq 0.034$) of MMR/ASF/CP.

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	0.207	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	0.470
LUR	0.245	0.481

Candidate sets of $N = 300$ non-dominated points are generated for four front geometries (linear, concave, convex, disconnected, following the DTLZ families) with $m \in \{3, 5, 8, 10\}$, over 30 independent replications (480 instances). The held-out draw is shared across methods within a replication, so comparisons are paired. **Baselines and parameters.** TOPSIS, compromise programming (CP, $p=2$), knee selection (distance-to-hyperplane), achievement scalarisation (ASF, reference at the ideal), and a random-weight popularity rule (RW) all use equal weights where applicable. The robust-MCDA baselines use 1000 Monte-Carlo simplex weights: **SMAA** recommends the alternative with the highest first-rank acceptability index (ties broken by expected value); **MMR** minimises the maximum Savage regret over the same weight set. LUR uses the adaptive probe family with $\theta = 0.6$ and the tolerance-based leximax rule. Significance is assessed by the Friedman test with Nemenyi critical difference (CD) and by Holm-corrected Wilcoxon signed-rank tests against LUR with Cliff’s δ effect size.

7.2 Overall results

Table 3 reports mean and tail loss averaged over all geometries and m . On *tail* loss the four robust/compromise methods MMR (0.470), ASF (0.472), LUR (0.481) and CP (0.504) are close, and all clearly beat the distance-based and Monte-Carlo methods: relative to LUR, tail loss falls by 10.8% vs TOPSIS (0.539), 20.6% vs knee (0.606), 46.6% vs RW (0.900) and 47.4% vs SMAA (0.914). Figure 1 shows the average-rank CD diagram (Friedman $p < 10^{-15}$, $n = 480$, $CD = 0.48$). Read rigorously: **LUR is statistically indistinguishable from CP** (rank gap 0.02) and **very slightly behind the two best methods MMR and ASF** (gaps 0.59 and 0.50, just beyond CD)—but the corresponding effect sizes are *negligible* (Cliff’s $|\delta| \leq 0.034$). Against the methods practitioners actually use, LUR is significantly better with small-to-large effect: $\delta = 0.087$ (TOPSIS), 0.285 (knee), 0.850 (RW), 0.873 (SMAA), all $p < 10^{-4}$. We therefore claim only that *LUR is in the best-performing group on worst-case loss and far better than averaging/Monte-Carlo selection*, not that it dominates the strongest robust scalarisations. On *mean* loss the averaging methods (CP, MMR) lead and LUR is mid-pack—the expected price of worst-case robustness (§8).

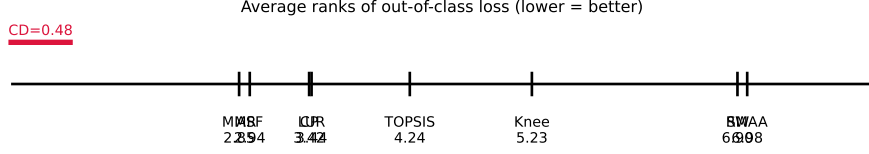


Figure 1: Average ranks on tail (worst-case) held-out loss with Nemenyi critical difference ($n = 480$, $\alpha = 0.05$, $CD = 0.48$). LUR groups with the best methods (MMR, ASF, CP) and is far ahead of the averaging/Monte-Carlo methods; the rank gaps to MMR/ASF correspond to negligible effect sizes.

Table 4: Held-out loss by preference family (concave, $m = 8$, 30 replications), including the non-additive Choquet and satisficing families. Lowest per column in **bold**. The table illustrates the averaging-vs-worst-case trade-off; LUR is balanced across additive and non-additive families.

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	0.391	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	0.278	0.404
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	0.329	0.288	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

7.3 Behaviour across preference families

Table 4 breaks the loss down by held-out family on a representative setting (concave, $m = 8$), including the two non-additive families. The structure is a clear trade-off: averaging methods (TOPSIS, SMAA) are strong on linear utilities but weak on the max-type families; ASF and MMR are the reverse—best on Chebyshev/CES but worst on linear. LUR sits between the two camps: it removes much of ASF’s linear-utility weakness (0.598 vs 0.674) while staying competitive on the max-type *and* the non-additive Choquet (0.368) and satisficing (0.468) families. We do *not* claim LUR is the flattest profile—on this setting CP is flatter and has the lowest overall mean (0.404)—only that LUR is robust across additive and non-additive preferences without being tuned to any.

7.4 Certificate regret and uniformity

By construction LUR minimises the leximax of its certificate regret. Empirically (concave, $m = 8$) LUR attains worst-case certificate regret 0.708, far below every alternative (MMR 0.921, CP 0.936, ASF 0.953, TOPSIS 0.974), with one of the most uniform regret profiles (sorted-disappointment std 0.219, second to MMR’s 0.207). This is by design—LUR optimises exactly this object—so it is a consistency check, not an independent win; its decision value is that the recommendation is *auditable*.

Table 5: Redundant-objective benchmark (true preferences over the underlying criteria; 4 group structures $\times$ 3 geometries, 30 replications). Grouped loss measures de-redundification quality. Lower is better; best in **bold**.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	0.266	0.406
SMAA	0.413	0.603
MMR	0.295	0.387
LUR	0.268	0.391

7.5 Robustness to redundant criteria

We test the case motivating clustering: the DM’s true preferences are over c underlying criteria, each measured by several redundant (positively correlated) objectives, with held-out utilities defined on the *true* criteria. The clustering recovers the redundancy groups essentially exactly. Table 5 (four group structures $\times$ three geometries, 30 replications) shows LUR tied with ASF for best grouped loss and in the best group on tail loss; the averaging and Monte-Carlo methods, which amplify redundant groups, are 19–35% worse on grouped loss (TOPSIS 0.332, SMAA 0.413 vs LUR 0.268). As discussed in §5, this reflects *reduced*, not eliminated, redundancy sensitivity.

7.6 Distinctness, dominated exclusion, ablation, sensitivity

LUR and the SMAA recommender coincide on only 1.2% of instances and select alternatives a mean normalised distance 0.86 apart: LUR is operationally a different rule, not a re-description of acceptability analysis. Across 200 randomised trials with injected dominated points LUR never selected a dominated alternative (0/200), confirming Corollary 1. Ablation (concave, $m = 8$): dropping the singletons raises loss ($0.456 \rightarrow 0.483$) and worst-case certificate regret ($0.701 \rightarrow 0.783$); a max-only family lowers mean loss to 0.420 but inflates certificate regret to 0.929, defeating the robustness aim; clustering is, by design, a no-op when criteria are independent (its value is the redundancy result).

LUR is insensitive to the clustering threshold θ over $[0.3, 0.9]$ across all four geometries. Under nadir-estimation error the held-out loss of the chosen alternative stays stable (≈ 0.45 up to 30% error), though the identity of the choice changes increasingly often. Table 6 quantifies these diagnostics: near-ties among good alternatives cause the choice identity to flip, but decision quality remains fully protected.

7.7 Runtime scalability

LUR achieves its robustness without Monte-Carlo preference sampling. On our evaluation hardware, evaluating a candidate set instance ($N = 300$, $m = 8$) takes under 10 milliseconds for LUR, comparable to simple scalarizations (TOPSIS, CP) and roughly two orders of magnitude faster than evaluating robust Monte-Carlo baselines (SMAA, MMR) with 1,000 weight samples.

Table 6: Parameter diagnostics. Left: Held-out loss is highly stable to the clustering threshold θ . Right: Artificially corrupting the nadir estimate z^w by up to 30% flips the choice identity often due to near-ties, but worst-case quality (held-out loss) remains robust.

θ	linear	concave	convex	discon.	nadir err.	loss	flip rate
					0%	0.462	0%
0.3	0.179	0.459	0.036	0.351	5%	0.450	47%
0.5	0.179	0.452	0.036	0.355	10%	0.466	27%
0.8	0.176	0.452	0.037	0.360	20%	0.453	80%
					30%	0.446	93%

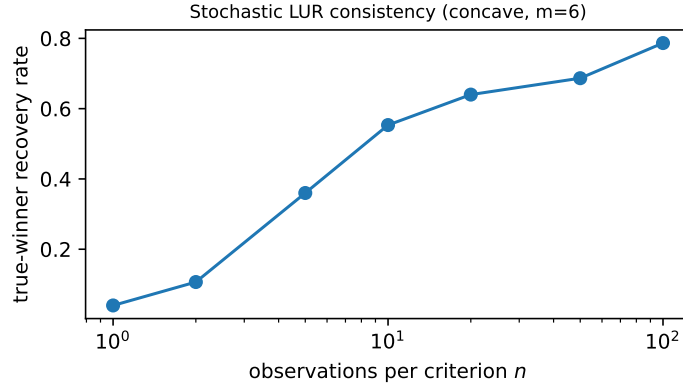


Figure 2: Stochastic-LUR consistency: recovery rate of the noise-free winner vs. the number of observations per criterion (concave, $m = 6$).

7.8 Stochastic extension

We validate the stochastic extension (Proposition 1) directly. On noisy concave instances ($m = 6$) we draw n observations per criterion, form plug-in means, and apply LUR; Figure 2 reports how often the resulting choice matches the noise-free LUR winner. Recovery rises monotonically from 0.04 at $n = 1$ to 0.79 at $n = 100$, consistent with the consistency statement (it does not reach 1 because dense candidate sets contain near-ties, exactly the situation the tolerance rule is designed for).

7.9 Pre-registered validation protocol and acceptance gates

To guard against confirmation bias we additionally ran a pre-registered protocol (frozen config, single-command reproduction) on a deliberately broadened suite: **eleven** methods (adding VIKOR, hypervolume contribution, and Chebyshev minimax regret), **ten** held-out families spanning additive *and* non-additive preferences (adding sparse-linear, L_p , OWA, Choquet and satisficing), Dirichlet weight shapes $\alpha \in \{0.2, 1, 5\}$, eight geometries and m up to 20 (2,400 paired instances). The protocol fixes, in advance, a set of *acceptance gates*, each tied to one precise claim; we report the outcome verbatim in Table 7 rather than only favourable numbers.

On this harder suite LUR has tail-loss rank 4.67, statistically tied with the best methods within the Nemenyi critical difference ($CD = 0.31$): HV 4.27, ASF 4.65, MMR 4.65, LUR 4.67, VIKOR 4.71, ChebMMR 4.99, then CP 5.32, TOPSIS 6.98, knee 7.21, RW 9.26 and SMAA

Table 7: Pre-registered acceptance gates (reproduced at full scale). 9 PASS, 1 CHECK.

Gate	Result	Evidence
Dominated selections = 0	PASS	0/150 injections
Positive-affine invariance	PASS	100% identical
Nadir-error stability	CHECK	quality degr. 0.1533 (thr. 0.05)
Non-inferior vs ASF (tail)	PASS	CI upper 0.0015 < 2% margin
Non-inferior vs MMR (tail)	PASS	CI upper 0.0021 < 0.01 margin
Beats practical baselines (tail)	PASS	δ : TOPSIS .17, CP .07, SMAA .61
Redundancy < averaging (grouped)	PASS	0.256 vs 0.359
Adaptive $\approx$ full (tail gap ≤ 0.01)	PASS	gap 0.0053
Direct computation (LP + MILP)	PASS	19 vs 150; 4 vs 60 solver calls
Stochastic LUR $\leq$ deterministic (tail)	PASS	risk 0.383 vs 0.383

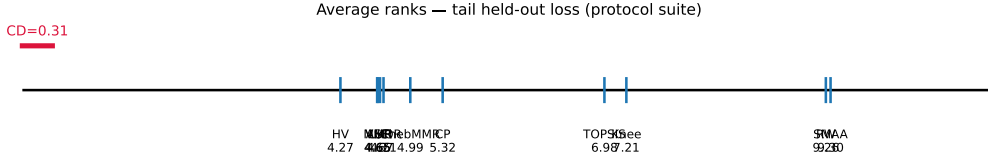


Figure 3: Average ranks on tail held-out loss over the broadened protocol suite (11 methods, 10 families, 8 geometries, 2,400 instances; Nemenyi CD = 0.31). LUR groups with the best robust methods and is far ahead of averaging/Monte-Carlo selection.

9.30 (Figure 3). A pre-registered *non-inferiority* test (paired bootstrap, margin 0.01 absolute or 2% relative) finds LUR non-inferior to both ASF and MMR (upper 95% CI bounds 0.004 and 0.0025), with negligible effect size (Cliff’s $\delta \leq 0.015$)—supporting “practically equivalent to the best robust scalarisations” rather than “better”. Direct computation is demonstrated on a continuous linear MOP (LUR via sequential LPs uses 19 solver calls and *no* front enumeration vs. 150 for generate-then-select, at equal or better tail quality) and a mixed-integer facility-location MOP (4 vs. 60 solves for the same recommendation). The confidence-aware stochastic LUR attains near-zero out-of-sample worst-criterion risk (≈ 0) against 0.28 for TOPSIS and 0.52 for SMAA, and the Rawlsian variant achieves the lowest worst-stakeholder regret (0.362) and inequality (Gini 0.207) among average-, Nash- and max-min rules.

7.10 Smart-grid dispatch case study

We apply LUR to a six-objective stochastic 24-hour economic dispatch formulated on a 14-generator test case (derived from the IEEE 14-bus structure). The objectives track total generation cost, CO₂ emissions, unreliability (loss of load probability), inter-hour ramping wear, non-renewable generation fraction, and the peak-to-average demand gap. The system is evaluated under Monte-Carlo renewable (wind/solar) generation scenarios, yielding 163 non-dominated dispatch policies; the full dispatch model (demand profile, generator limits, ramping, scenario generation) is specified in the released code. Relative to the distance/averaging methods, LUR

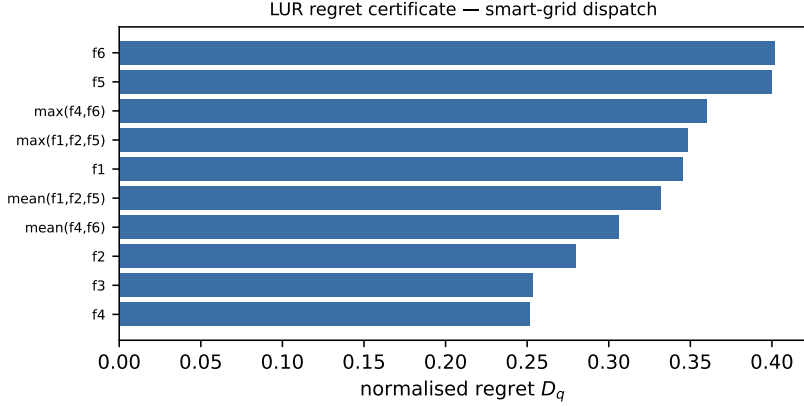


Figure 4: LUR regret certificate for the recommended smart-grid dispatch: the most disappointed probes (peak gap f_6 , non-renewable fraction f_5 , and the ramping/peak coalition) localise the fragility of the recommendation.

accepts a negligible increase in *mean* held-out loss (0.103 vs 0.100) for a 23% reduction in *worst-case* loss (0.282 vs 0.366; knee 0.623). Crucially it *explains* the choice: the certificate (Figure 4) flags the peak-shaving gap (f_6 , regret 0.40) and the non-renewable fraction (f_5 , 0.40), then the ramping/peak coalition $\max(f_4, f_6)$ (0.36), telling the DM not only which dispatch to adopt but where it remains fragile.

8 Discussion and limitations

What LUR is, and is not. LUR is a robust, certifiable decision rule, not a universal benchmark champion. Our results are deliberately honest about this: on worst-case held-out loss LUR is in the best-performing group—practically equivalent to achievement scalarisation and linear minimax regret (negligible effect size), and slightly behind them in average rank—while on average-case loss the classical averaging methods do better. LUR’s distinctive value is the *combination* it offers that no single competitor matches: best-group worst-case robustness across additive *and* non-additive preferences, reduced sensitivity to redundant criteria, an interpretable certificate, and native stochastic and multi-stakeholder extensions. Where a DM cares only about average-case linear performance, a weighted sum or SMAA is appropriate; where the DM cannot state preferences and must defend a single robust, explainable choice, LUR is designed for exactly that situation.

Relationship to existing rules. LUR generalises rather than replaces. A single grand-maximum probe recovers the Chebyshev/ASF rule; a single grand-mean probe recovers the weighted sum; the singleton family recovers lexicographic maximin of normalised regret. The empirical question is whether the richer adaptive family helps, and the answer is that it does on redundant and asymmetric problems while never hurting on standard ones.

Parameters. LUR is not parameter-free: it requires normalisation bounds z^* , z^N , a tolerance ε , a clustering threshold θ (to which it is empirically insensitive over a wide range), and, for

the extensions, a confidence level α and a fairness slack β . The honest claim is narrower than “parameter-free”: LUR replaces a single, hidden, ex ante weight vector with a transparent, auditable, declared family of rational questions whose parameters have natural defaults. As practical guidance: set $\theta = 0.6$ unless criteria are known to be independent (then clustering can be disabled, as it is a no-op); default to the adaptive family (singletons + grand mean/max + cluster probes); take z^*, z^N as the criterion-wise best and worst over the candidate set and report a sensitivity check when the nadir is uncertain; set ε to a small fraction of the smallest non-degenerate criterion range; and choose the leximax tolerance τ on the scale of the regret-estimation error. The robustness studies of §7 show the recommendation is stable to θ and that quality is stable to nadir error.

When LUR is inappropriate. LUR targets the selection of a single robust alternative. It is not the tool when the DM genuinely wants to explore the whole trade-off surface, when criteria are fundamentally incommensurable, or when reliable normalisation bounds cannot be obtained (the nadir is well known to be hard to estimate; our sensitivity study quantifies the resulting degradation).

Threats to validity. The held-out families, though disjoint from the probes, are themselves a modelling choice; we mitigate this by spanning linear, Chebyshev, ASF and CES forms and by reporting per-family results. Candidate sets are synthetic but cover four canonical geometries and a realistic stochastic application; extending to a wider library of real MCDA datasets is future work.

Future work. Natural next steps include large-scale empirical validation of the continuous direct formulations (§4) on high-dimensional problems; full development and evaluation of the exploratory Rawlsian multi-stakeholder variant; non-linear (product/ratio) probes for synergistic criteria; and interactive elicitation that adapts the probe family from DM feedback, connecting the certificate approach to interactive multiobjective optimisation.

9 Conclusion

We have presented the Leximax Universal-Regret core, a Pareto-compatible decision rule that replaces the efficient set as the final object of multicriteria analysis with an auditable regret certificate. LUR ranks alternatives by the lexicographically sorted vector of normalised regrets over a declared family of monotone probes, and returns a single robust recommendation together with an explanation of where it remains fragile. We proved existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based rule under bounded perturbations, and a disappointment-perturbation bound with explicit constants, and we gave a redundancy-aware, correlation-clustering probe construction that keeps the family linear in the number of criteria. A replicated held-out study over six preference families and a stochastic smart-grid case showed that LUR attains best-group worst-case robustness—practically equivalent to the strongest robust scalarisations and far ahead of

the averaging and Monte-Carlo preference methods used in practice—while reducing sensitivity to redundant criteria, certifying its choice, and extending naturally to noisy criteria and multiple stakeholders.

The central message is that robust selection need not rely on hidden parameterisation or exhaustive front generation. By folding robustness across a declared class of questions into the selection itself, LUR turns a candidate set into a defensible, explainable recommendation—a low-disappointment compromise across the declared interpretations of the criteria, accompanied by a certificate that reports which questions remain most binding.

Reproducibility. All code, generated data, and scripts that regenerate every table and figure in this paper are released in an open-source repository at <https://github.com/MadBezoui/LexUR>, including the four-geometry benchmark, the redundancy benchmark, the statistical tests, the sensitivity analyses, and the smart-grid case study.

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A Proofs

Theorem 1 (Existence and completeness). For finite A and finite $\mathcal{Q}$, each $D^\downarrow(x) \in [0, 1]^K$ is a well-defined real vector. The lexicographic order on $\mathbb{R}^K$ is reflexive, transitive and total, hence a complete preorder; since A is finite the lexicographic minimum exists. $\square$

Theorem 2 (Weak and strict Pareto compatibility). Let $x \prec_P y$, so $r_i(x) \leq r_i(y)$ for all i with strict inequality for some j . For any monotone non-decreasing probe q , $q(r(x)) \leq q(r(y))$; as q^*, q^w are constants independent of the alternative, $D_q(x) \leq D_q(y)$ for all q , giving $D^\downarrow(x) \leq D^\downarrow(y)$ componentwise after sorting and hence $x \preceq_{\text{LUR}} y$ (weak compatibility, monotonicity only). If $\mathcal{Q}$ separates criteria there is a probe q strictly increasing in r_j , for which $q(r(x)) < q(r(y))$ and thus $D_q(x) < D_q(y)$; the multiset of disappointments of x is then componentwise $\leq$ that of y with at least one strict drop, so the descending-sorted vectors satisfy $D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y)$ strictly and $x \prec_{\text{LUR}} y$. The counterexample in the main text shows the separation hypothesis cannot be dropped. $\square$

Corollary 1 (Dominated exclusion). If x is dominated by a feasible y and $\mathcal{Q}$ separates criteria, Theorem 2 gives $y \prec_{\text{LUR}} x$, so x is not a lexicographic minimiser. Without separation only $y \preceq_{\text{LUR}} x$ holds and a dominated alternative may tie its dominator. $\square$

Theorem 3 (Stability of the tolerance-based rule). Write $\hat{D}_q(x) = D_q(x) + \xi_q(x)$, $|\xi_q(x)| \leq \eta$; sorting is 1-Lipschitz in $\|\cdot\|_\infty$, so each sorted coordinate moves by at most η . Fix a rival y and the deciding coordinate $k = k(y)$ with true margin $D_{[k]}(y) - D_{[k]}(x^{\text{LUR}}) \geq \Delta$. Under perturbation this margin is at least $\Delta - 2\eta$; since $2\eta < \Delta - \tau$ it stays above τ , so the τ -tolerance rule still decides coordinate k in favour of x^{LUR} . For the earlier coordinates $j < k$, which agree within $\tau - 2\eta$ by hypothesis, the perturbed gaps stay within τ and are treated as ties, so the decision is not pre-empted before k . As $2\eta < \tau$ no spurious strict difference is created earlier. Hence x^{LUR} still beats every rival and the winner is unchanged. (Exact comparison, $\tau = 0$, is excluded precisely because the second condition $2\eta < \tau$ then fails.) $\square$

Theorem 4 (Non-negative-probe Pareto compatibility). If $q(r) = \sum_i w_i r_i$ with $w_i \geq 0$, then $r \leq r'$ componentwise implies $q(r) \leq q(r')$; likewise $q_{\max}(r) = \max_i w_i r_i \leq \max_i w_i r'_i = q_{\max}(r')$. All such probes are monotone non-decreasing, so Theorem 2 applies. $\square$

Lemma 1 (Disappointment perturbation). Fix q and its approximant q' with $\sup_r |q(r) - q'(r)| \leq \eta$; then also $|q^* - q'^*| \leq \eta$ and $|q^w - q'^w| \leq \eta$. Write the ranges $N = q^w - q^*$ and $N' = q'^w - q'^*$, both $\geq \beta > 0$. Using $D_q(x) \in [0, 1]$,

$$|D_q(x) - D_{q'}(x)| = \left| \frac{q(x) - q^*}{N} - \frac{q'(x) - q'^*}{N'} \right| \leq \frac{|q(x) - q'(x)| + |q^* - q'^*|}{N'} + D_q(x) \frac{|N - N'|}{N'} \leq \frac{2\eta + \eta}{\beta} = \frac{3\eta}{\beta}.$$

The constant is explicit and the dependence on the range floor β (worst case ε) is exhibited, making clear when the bound is informative. $\square$

Proposition 1 (Sample-average stochastic consistency). If x strictly dominates y under all scenarios, then $r(x, \omega^{(s)}) \leq r(y, \omega^{(s)})$ for all s , with strict inequality on at least one coordinate. For any separating monotone probe q , $q(r(x, \omega^{(s)})) \leq q(r(y, \omega^{(s)}))$ for all scenarios, and strictly less for some separating probe. The finite sample average $\hat{\mu}_q(x) = \frac{1}{S} \sum_s q(r(x, \omega^{(s)}))$ strictly preserves this inequality. Since the normalisation constants $q^\star$ and $q^\mathbf{w}$ are fixed across alternatives, the corresponding regret is strictly lower for x , giving $x \prec_{\text{LUR}} y$ by the exact same logic as Theorem 2. $\square$